COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 007	**SECTION:**	CPE11S4
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NAME:	Mangana, Prince Adrian M.	**INSTRUCTOR:**	Robin Valenzuela

ACTIVITY NO. 3.1 Control Structures

1. PROCEDURE OUTPUT

Perform each of these steps for the problems below:

1. Develop a C++ program that will determine if a department store customer has exceeded the credit limit on a charge account. For each customer, the following facts are available:
    a. Account number
    b. Balance at the beginning of the month
    c. Total of all items charged by this customer this month
    d. Total of all credits applied to this customer's account this month

ifelse.cpp | grades.cpp | menu.cpp | grades.exe | balance.cpp

balance.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main() {
    int AccountNumber;
    double BeginningBalance, TotalCharges, TotalCredits, CreditLimit, NewBalance;

        cout << "Enter Account Number: ";
        cin >> AccountNumber;

        cout << "Enter Beginning Balance: ";
        cin >> BeginningBalance;

        cout << "Enter Total Charges: ";
        cin >> TotalCharges;

        cout << "Enter Total Credits: ";
        cin >> TotalCredits;

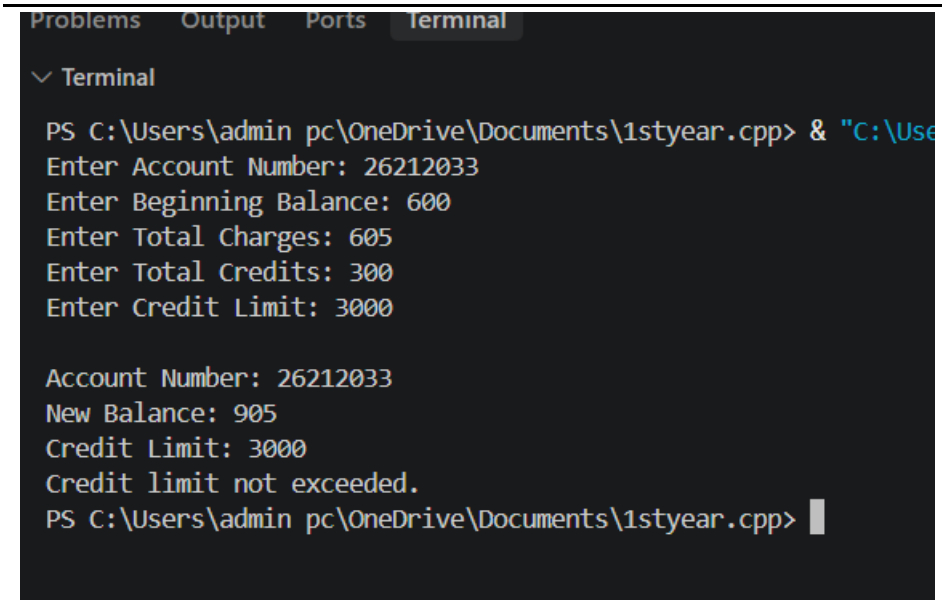
        cout << "Enter Credit Limit: ";
        cin >> CreditLimit;


            NewBalance = BeginningBalance + TotalCharges - TotalCredits;

                cout << "\nAccount Number: " << AccountNumber << endl;
                cout << "New Balance: " << NewBalance << endl;
                cout << "Credit Limit: " << CreditLimit << endl;


        if (NewBalance > CreditLimit) {
            cout << "Credit limit exceeded!" << endl;
        } else {
            cout << "Credit limit not exceeded." << endl;
    }

    return 0;
}
```



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Because of the price of gasoline, drivers are concerned with the mileage obtained by their automobiles. One driver has kept track of several tankfuls of gasoline by recording miles driven and gallons used for each tankful. Develop a program that will input the miles driven and gallons used for each tankful. The program should calculate and display the miles per gallon obtained for each tankful. After processing all input information, the program should calculate and print the combined miles per gallon obtained for all tankfuls.

Gasoline.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main () {
    double MilesDriven, GallonsUsed, mpg, TotalMiles = 0, TotalGallons = 0, Combinedmpg;

        cout << "Enter Miles Driven (-1 to stop): ";
        cin >> MilesDriven;

        while (MilesDriven != -1) {
            cout << "Enter Gallons Used: ";
            cin >> GallonsUsed;

            Combinedmpg = MilesDriven / GallonsUsed;
            cout << "Miles per Gallon for this tankful: " << Combinedmpg << endl;

            TotalMiles = TotalMiles + MilesDriven;
            TotalGallons = TotalGallons + GallonsUsed;

            cout << "\nEnter Miles Driven (-1 to stop): ";
            cin >> MilesDriven;
        }

        if (TotalGallons !=0) {
            Combinedmpg = TotalMiles / TotalGallons;

            cout << "\nCombined Miles per Gallon: " << Combinedmpg << endl;

        } else {
            cout << "\nNo data Entered." << endl;
        }

        return 0;
}
```

```
PS C:\Users\admin pc\OneDrive\Documents\1styear.cpp> & "C:\Use
Enter Miles Driven (-1 to stop): 65.25
Enter Gallons Used: 32.54
Miles per Gallon for this tankful: 2.00522

Enter Miles Driven (-1 to stop): 75
Enter Gallons Used: 16.45
Miles per Gallon for this tankful: 4.55927

Enter Miles Driven (-1 to stop): 86.4
Enter Gallons Used: 41.5
Miles per Gallon for this tankful: 2.08193

Enter Miles Driven (-1 to stop): -1

Combined Miles per Gallon: 2.5047
PS C:\Users\admin pc\OneDrive\Documents\1styear.cpp>
```

2. Create a program that will calculate the cost of sending a small parcel. The post office charges P5.00 for the first 300g, and P2.00 for every 100g thereafter (rounded up), up to a maximum weight of 1000g.

parcel.cpp > ...

```cpp
#include <iostream>
using namespace std;

int main() {
    int weight, extraWeight, extraUnits;
    double cost;

    cout << "Enter parcel weight (g): ";
    cin >> weight;

    if (weight <= 0) {
        cout << "Invalid weight!";
    }
    else if (weight > 1000) {
        cout << "Parcel exceeds the maximum weight of 1000g.";
    }
    else if (weight <= 300) {
        cost = 5.00;
        cout << "Shipping Cost: P" << cost;
    }
    else {
        extraWeight = weight - 300;

        extraUnits = (extraWeight + 99) / 100;

        cost = 5.00 + (extraUnits * 2);

        cout << "Shipping Cost: P" << cost;
    }

    return 0;
}
```

```
PS C:\Users\admin pc\OneDrive\Documents\1styear.cpp> & "C:\Use
Enter parcel weight (g): 456
Shipping Cost: P9
PS C:\Users\admin pc\OneDrive\Documents\1styear.cpp> & "C:\Use
Enter parcel weight (g): 5256
Parcel exceeds the maximum weight of 1000g.
PS C:\Users\admin pc\OneDrive\Documents\1styear.cpp>
```

3. Write a program that displays a menu for simple conversions such as the following:
(1) cm - inches
(2) inches - cm
(3) feet - meter
(4) meter - feet
Once selected, the user will be asked to enter a float to be converted. After the conversion, the user would

be asked to convert another until the user doesn't want any more. Display your samples for all conversions

newmenu.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main() {
    float cm, inch, feet, meter;
    int choice;
    char answer;

    do {

        cout << "\nWhat did you want to convert?"
             << "\n1. Centimeter to Inch"
             << "\n2. Inch to Centimeter"
             << "\n3. Feet to Meter"
             << "\n4. Meter to Feet"
             << "\n5. Exit"
             << "\nEnter your choice: ";
        cin >> choice;

        switch (choice) {
            case 1:
                cout << "Enter value in Centimeter: ";
                cin >> cm;
                inch = cm / 2.54;
                cout << cm << " Centimeter = " << inch << " Inch" << endl;
                break;

            case 2:
                cout << "Enter value in Inch: ";
                cin >> inch;
                cm = inch * 2.54;
                cout << inch << " Inch = " << cm << " Centimeter" << endl;
                break;

            case 3:
                cout << "Enter value in Feet: ";
                cin >> feet;
                meter = feet * 0.3048;
                cout << feet << " Feet = " << meter << " Meter" << endl;
                break;

            case 4:
                cout << "Enter value in Meter: ";
                cin >> meter;
                feet = meter / 0.3048;
                cout << meter << " Meter = " << feet << " Feet" << endl;
                break;

            case 5:
                cout << "Exiting the program." << endl;
                answer = 'N';
                continue;

            default:
                cout << "Invalid choice! Please try again." << endl;
        }

        cout << "\nDo you want to perform another conversion? (Y/N): ";
        cin >> answer;

    } while (answer == 'Y' || answer == 'y');

    cout << "\nExiting the program." << endl;

    return 0;
}
```

```
What did you want to convert?
1. Centimeter to Inch
2. Inch to Centimeter
3. Feet to Meter
4. Meter to Feet
5. Exit
Enter your choice: 1
Enter value in Centimeter: 2.36
2.36 Centimeter = 0.929134 Inch

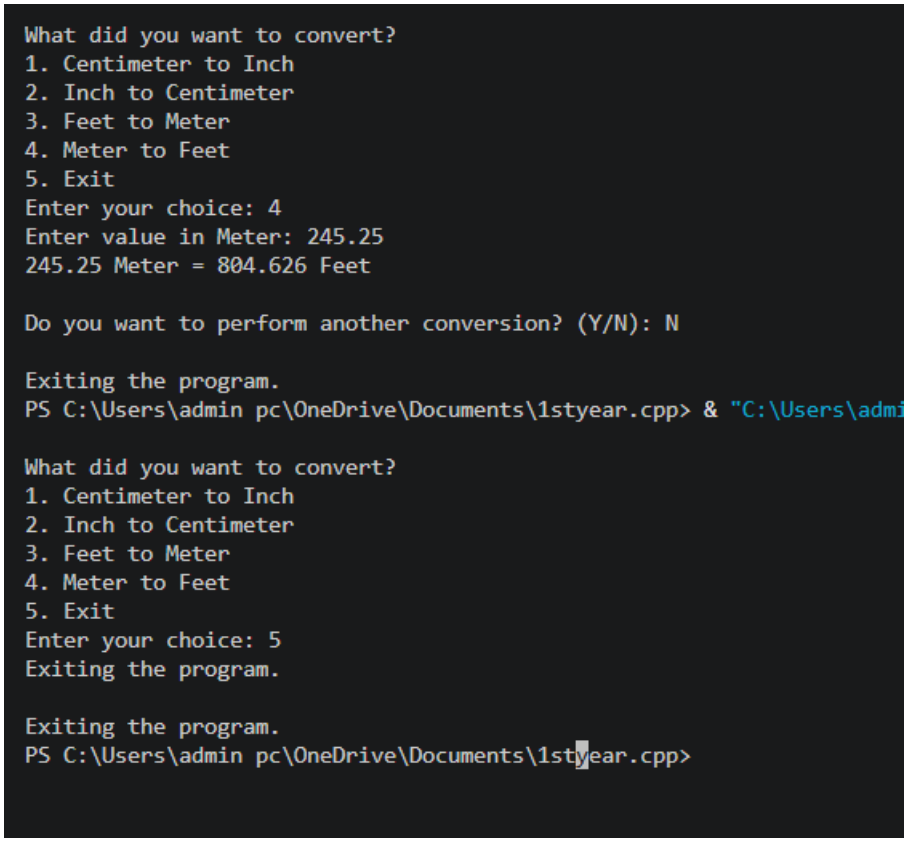
Do you want to perform another conversion? (Y/N): Y

What did you want to convert?
1. Centimeter to Inch
2. Inch to Centimeter
3. Feet to Meter
4. Meter to Feet
5. Exit
Enter your choice: 2
Enter value in Inch: 11.5
11.5 Inch = 29.21 Centimeter

Do you want to perform another conversion? (Y/N): y

What did you want to convert?
1. Centimeter to Inch
2. Inch to Centimeter
3. Feet to Meter
4. Meter to Feet
5. Exit
Enter your choice: 3
Enter value in Feet: 6.7
6.7 Feet = 2.04216 Meter

Do you want to perform another conversion? (Y/N): Y
```



4. Write a program that displays a menu for simple computation of formula such as the following:
(1) Area of circle, will ask for radius from user
(2) Area of rectangle, will ask for L and W
(3) Area of triangle, will ask for B and H
(4) Area of square - feet, will ask for S
Once selected, the user will be asked to enter a float and be computed as selected. After the computation, the user would be asked to compute another until the user doesn't want anymore. Display your samples for all conversions

```cpp
#include <iostream>
using namespace std;

int main() {
    float radius, length, width, base, height, side, area;
    int choice;
    char answer = 'Y';

    do {

        cout << "\nWhat Area did you want to calculate?"
             << "\n1. Area of Circle"
             << "\n2. Area of Rectangle"
             << "\n3. Area of Triangle"
             << "\n4. Area of Square"
             << "\n5. Exit"
             << "\nEnter your choice: ";
        cin >> choice;

        switch (choice) {
            case 1:
                cout << "Enter Radius: ";
                cin >> radius;
                area = 3.14159 * radius * radius;
                cout << "Area of Circle: " << area << endl;
                break;

            case 2:
                cout << "Enter the Length: ";
                cin >> length;
                cout << "Enter the Width: ";
                cin >> width;
                area = length * width;
                cout << "Area of Rectangle: " << area << endl;
                break;

            case 3:
                cout << "Enter the Base: ";
                cin >> base;
                cout << "Enter the Height: ";
                cin >> height;
                area = 0.5 * base * height;
                cout << "Area of Triangle: " << area << endl;
                break;
```

```
            break;

        case 4:
            cout << "Enter the Side: ";
            cin >> side;
            area = side * side;
            cout << "Area of Square: " << area << endl;
            break;

        case 5:
            cout << "\nExiting the program." << endl;
            return 0;

        default:
            cout << "\nInvalid choice! Please try again." << endl;
            continue;
        }

        cout << "\nDo you want to perform another calculation? (Y/N): ";
        cin >> answer;

    } while (answer == 'Y' || answer == 'y');

    cout << "Exiting the program..." << endl;

    return 0;
}
```

```
What Area did you want to calculate?
1. Area of Circle
2. Area of Rectangle
3. Area of Triangle
4. Area of Square
5. Exit
Enter your choice: 1
Enter Radius: 1.56
Area of Circle: 7.64537

Do you want to perform another calculation? (Y/N): y

What Area did you want to calculate?
1. Area of Circle
2. Area of Rectangle
3. Area of Triangle
4. Area of Square
5. Exit
Enter your choice: 2
Enter the Length: 5
Enter the Width: 8
Area of Rectangle: 40

Do you want to perform another calculation? (Y/N): Y

What Area did you want to calculate?
1. Area of Circle
2. Area of Rectangle
3. Area of Triangle
4. Area of Square
5. Exit
Enter your choice: 3
Enter your choice: 3
Enter the Base: 8.26
Enter the Height: 11.25
Area of Triangle: 46.4625

Do you want to perform another calculation? (Y/N): Y

What Area did you want to calculate?
1. Area of Circle
2. Area of Rectangle
3. Area of Triangle
4. Area of Square
5. Exit
Enter your choice: 4
Enter the Side: 9.65
Area of Square: 93.1225

Do you want to perform another calculation? (Y/N): n
Exiting the program...
PS C:\Users\admin pc\OneDrive\Documents\1styear.cpp> & "

What Area did you want to calculate?
1. Area of Circle
2. Area of Rectangle
3. Area of Triangle
4. Area of Square
5. Exit
Enter your choice: 5

Exiting the program.
```

```
What Area did you want to calculate?
1. Area of Circle
2. Area of Rectangle
3. Area of Triangle
4. Area of Square
5. Exit
Enter your choice: 7

Invalid choice! Please try again.

What Area did you want to calculate?
1. Area of Circle
2. Area of Rectangle
3. Area of Triangle
4. Area of Square
5. Exit
Enter your choice: 5

Exiting the program.
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

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Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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